Supplementary Information: Quantum-inspired catalytic QUBO search for balanced hypergraph partitioning

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The main manuscript defines the cut objectives and QUBO encoding. Here, classical baselines lead into balance correlations and coordinate barriers. The catalytic response then motivates the shared solver policy. Mechanism controls distinguish dynamical effects from classical spectral information and document failures. Computational costs delimit practical usefulness, followed by an audit of the evidence. Primary benchmarks and exploratory controls retain their specified designs and are not pooled. In cut/imbalance tables, parentheses give observed imbalance in percent. Historical cut/time tables mark the parenthesized runtime with s. The allowed imbalance is an upper bound, not a required imbalance.

Supplementary Note 1. Classical baselines

Multilevel search

KaHyPar optimizes the native hypergraph objective through a hierarchy of smaller problems and subsequent refinement[1]. Its preprocessing identifies communities[2] and, when the configuration's thresholds are met, applies min-hash sparsification. Coarsening contracts strongly connected vertices while tracking their combined load. This retains a coarse partitioning problem rather than replacing each net by its full clique. On the smallest hypergraph, a pool of initial partitioners finds candidate balanced assignments. Uncoarsening then lifts the selected assignment through the hierarchy and refines it at each level.

The cut-oriented configurations used here select multilevel coarsening with heavy-edge, community-aware ratings. Their initial partitioning stage uses recursive multi-level partitioning and 20 pool trials, even though the outer partitioning mode is direct. Refinement combines Fiduccia–Mattheyses (FM) gain-based vertex moves[3] with HyperFlowCutter[4]. Flow refinement optimizes a region between block pairs through max-flow/min-cut computations[5]; it can change several vertex assignments together. The classical reference thus combines multiple search scales and discrete collective moves. Its quality is not explained solely by changing an optimizer for a fixed continuous objective.

Reference protocol

The current main tables use three direct single-call KaHyPar-K trials per input and capacity, with seeds 0–2, one CPU thread and independently scored returned partitions. KaHyPar does not provide the same exact-balance search guarantee as its usual imbalance-tolerant configurations. Direct strict calls therefore constitute a

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limited comparator; allowing small slack and then repairing and refining to exact balance can produce substantially lower cuts. The main win count does not establish an advantage over those stronger classical pipelines. Historical references below retain their recorded repeated-search protocols and are separate from the current single-call timing cohort.

Every reference receives the original hypergraph, the cut-net objective, two blocks and the requested integer capacity. Both ordinary and evolutionary runs use cut-oriented multilevel refinement. Ordinary time-limited repeated partitioning repeats independent multilevel searches. KaHyPar-E instead maintains and evolves a population of partitions[6]; the two modes are evaluated separately. Each result includes the seed, nominal time allowance, actual elapsed time and returned partition. The single-run enumeration controls, historical strict-balance references and matched 1%/3% runs retain their own recorded budgets; they are not presented as one uniform KaHyPar timing experiment.

Each process uses one CPU thread. Native cut and both block sizes are recomputed from its saved partition, rather than inferred from the reported objective alone. An output that violates the capacity is not a feasible comparator. Measured elapsed time is retained because the nominal limit is soft. The hardware and concurrency are specified in Supplementary Note 6. Reference partitions serve only for comparison; they never initialize FEM trajectories.

Spectral readout

A second classical control tests how much quality is already present in the incidence modes used to construct collective catalysts. Hypergraph Laplacians and spectral partitioning provide the structural background[7,8]; our readout and catalyst controls compare two uses of those same continuous directions. Each of the eight modes is stably sorted, and exactly $\lfloor n/2 \rfloor$ vertices are assigned to one block. The resulting partitions are scored on the original nets. This readout is exactly balanced and requires no FEM trajectory. It is therefore a control for classical structural information, not an additional FEM update or input partition. Its quality and charged construction time are compared with the catalytic runs in the mechanism and cost sections below.

Supplementary Note 2. Balance and dynamics

The classical references enforce capacity during discrete partitioning. For a variational binary-energy search, a quadratic balance penalty must also be evaluated while the labels remain uncertain. The same marginals can conceal different count fluctuations, and a correct binary endpoint can still obstruct coordinate descent. These are separate questions from whether the encoded cut matches the original hypergraph.

Constraint correlations

A faithful binary target still leaves a distributional question: how is balance represented before spins become discrete?

Let $p_i = \Pr(x_i = 1)$. These marginals do not specify the fluctuations of the vertex count. This distinction between marginal and joint trial distributions is central to variational inference[9]. For $N = \sum_i x_i$, $\mathrm{Var}(N) = \sum_i p_i(1 - p_i) + 2\sum_{i<j} \mathrm{Cov}(x_i, x_j)$. At $p_i = 1/2$, independent vertices have variance $n/4$. A uniform fixed-cardinality distribution has the same marginals but zero count variance, with off-diagonal covariance $-1/[4(n-1)]$ for even n. Exact enumeration on example B gives expected balance costs 24 and 0, respectively, at penalty 8. Including each distribution's own entropy at temperature 1 gives free energies 23.8307 and 1.7622. Both trial families use the same cut-net QUBO and deterministic minimum-energy net auxiliaries. The correlated distribution has a higher expected cut, 8.5909 versus 8.1484, so this comparison isolates feasibility correlations rather than improved optimization.

Cardinality-conditioned or cluster trial distributions could retain these correlations, as in richer cluster trial families[10]. Their scalable evaluation and unbiased readout remain untested here.

Slack entropy and count fluctuations

The band construction in Methods can represent a slack value more than once. Summing the Gibbs probability over all binary assignments with value

s gives multiplicity g_s; replacing these representations by one uniformly weighted integer would change the entropy. The independent vertex count and joint slack factor obey

$$\mathbb{E}[(N - L - S)^2] = (\mathbb{E}N - L - \mathbb{E}S)^2 + \sum_i p_i(1 - p_i) + \mathrm{Var}(S). \tag{1}$$

Only the first term depends on the slack mean. Minimizing over its binary representations therefore gives the multiplicity-weighted conditional distribution used in Methods; the variance terms remain in the energy. For an oriented strict target, the slack variance is absent. Centering the odd strict target at $n/2$ subtracts the constant $1/4$, without changing the fluctuation contribution.

Coordinate barriers

Quadratic constraint penalties and auxiliary elimination are standard ingredients of Ising and pseudo-Boolean formulations[11,12]. The bound below is our encoding-specific coordinate-barrier statement, checked independently on retained states.

For an oriented count error d, correcting d vertices raises native cut by at most $dD_{\max}$ and removes penalty λd^2. The endpoint therefore has feasible global minima when $\lambda > D_{\max}$. At a feasible state, however, one vertex flip incurs penalty λ and changes native cut by at most D_i, giving the main-text bound. Conditional binary auxiliary minimization attains the least encoded net cost, so holding those auxiliaries fixed cannot improve that bound. At $N = L + S$, flipping a slack bit of weight a_j similarly costs λa_j^2; a vertex flip pays at least $\lambda - D_i$. For odd centered balance, moves between the two legal counts are excluded from this obstruction, as detailed below.

These statements concern individual encoded-bit moves. A forbidden interval of overlaps between low-cost states is a different global property, addressed in the scope discussion of Supplementary Note 3.

The main-text coordinate-barrier lemma is checked on all 384 selected complete-QUBO states from the six-arm *finan512*/*thermal1* refinement study. Every state is exactly balanced and auxiliary-minimized, and every single-bit energy delta is nonnegative. *finan512* has $D_{\max} = 55$ and $\lambda = 56$; its smallest observed vertex-flip energy increase is 54. *thermal1* has $D_{\max} = 11$ and $\lambda = 12$; its smallest increase is 10. Already feasible raw endpoints (119 and 74, respectively) receive no vertex flips during subsequent descent. The smallest checked cuts, 438 and 2,578, still exceed strict KaHyPar cuts 169 and 291.

Holding balance aside, conditional auxiliary minimization permits some single-vertex native-cut decreases: median counts of such vertices are 18 on *finan512* and 138 on *thermal1* across the 192 states each. Those moves violate exact balance and are not applied. Thus, absence of an improving complete-QUBO one-bit move is not evidence of good partition quality. Independent finite differences check one most-favorable vertex per batch, and weighted small-instance enumeration checks every variable of every balanced assignment. This diagnostic neither updates the partitions nor uses KaHyPar states as initial conditions.

Block-label symmetry at strict capacity

The preceding coordinate-barrier lemma uses the oriented equality penalty. For odd n, put $b = \lfloor n/2 \rfloor$. The alternative $P_{\mathrm{sym}} = \lambda(N - b)(N - b - 1)$ vanishes at both legal counts. Complementing vertices and exchanging the net auxiliaries, $(\mathbf{x}, \mathbf{u}, \mathbf{v}) \mapsto (\mathbf{1} - \mathbf{x}, \mathbf{v}, \mathbf{u})$, preserves the entire binary energy. The original equality formulation remains exact for the unlabelled optimum because each feasible partition has an orientation with $N = b$; the new formulation removes its arbitrary label preference.

At $N = b$, a $0 \to 1$ flip has zero balance cost, while a $1 \to 0$ flip costs 2λ. At $N = b + 1$ these roles reverse. Hence some feasible one-bit moves are available, although the native objective can still have local minima. For even n the penalties coincide. For odd n their expectations differ by $\lambda(\sum_i p_i - b)$: the gradient difference is uniform and their Hessians in probability coordinates coincide. Exhaustive weighted small-instance checks verify the ground states, complement symmetry and single-bit deltas; independent derivatives verify the linear-field difference and unchanged Hessian. These statements do not predict a global convergence rate or a catalyst-specific benefit.

Supplementary Note 3. Catalysts in the FEM free energy

This note derives how the temporary interactions enter the classical free-energy machine (FEM) used in the main text. Temporary interactions in quantum annealing motivate modifying the intermediate energy while restoring the endpoint[13,14]. Here this principle is implemented in a variational objective over independent vertex labels and joint net-auxiliary blocks. We specify the incidence weights and mode construction, derive their free-energy derivatives, and finally connect those derivatives to the actual FEM update. The stability calculations use a fixed strict count target. Supplementary Note 4 specifies the complete search schedule and replica allocation.

Incidence weights for the local catalysts

The local pair catalyst couples vertices that share an original net. Its binary energy is the pair interaction defined in the main text, with positive node scales z_i and net coefficients a_e. We first use $a_e^{(0)} = 1/\max(1, \lfloor r_e^2/4 \rfloor)$, so a net's largest possible unscaled pair-cut contribution is bounded by its weight. The node scales then redistribute this coupling according to incidence degree and local degree contrast. They take the form $z_i = (1 + D_i)^{-\alpha} \exp(\xi\sigma_i)$, where σ_i is a bounded structural signal and α, ξ are fixed by the replica profile.

The structural signal describes the input rather than a candidate partition. The degree signal starts from $\ln(1 + D_i)$; the contrast signal subtracts a weighted mean of neighbouring log degrees, counting repeated co-occurrences in different nets. Signals are centered, standardized, clipped to two standard deviations, recentered and bounded. The final signals have absolute value at most one; vertices without the relevant incidence receive zero signal. After applying the node scales, a common positive rescaling of $a_e^{(0)}$ keeps $\sum_e w_e a_e \sum_{i<j\in e} z_i z_j$ equal to its unscaled value. This holds total pair coupling fixed while its spatial distribution changes.

The auxiliary coefficient is a temporary replica-dependent strength multiplied by $w_e \max(r_e/2 - 1, 0)$. It lowers states $(1,0)$ and $(0,1)$ without changing $(0,0)$ or $(1,1)$.

Local interaction derivatives under the FEM distribution

The pair term acts on the FEM probabilities through its exact Bernoulli expectation. Independence gives

$$\mathbb{E}_q[C_{\text{pair}}] = \sum_e w_e a_e \sum_{i<j\in e} z_i z_j (p_i + p_j - 2p_i p_j). \tag{2}$$

Writing $m_i = 2p_i - 1$, its derivative with respect to p_i is

$$\begin{aligned} g_i^{\text{pair}} = z_i^2 m_i \sum_{e\ni i} w_e a_e \\ - z_i \sum_{e\ni i} w_e a_e \sum_{j\in e} z_j m_j. \end{aligned} \tag{3}$$

The first term removes the diagonal self-interaction included in the second sum. Net-to-vertex and vertex-to-net incidence products evaluate both terms without constructing the expanded clique. The resulting off-diagonal interaction changes probability curvature according to the incidence structure, rather than introducing a replacement cut objective at the endpoint.

The corresponding probability Hessian has zero diagonal and entries $-2\gamma \sum_{e\ni i,j} w_e a_e z_i z_j$ off the diagonal. All coefficients are held fixed in this derivative.

The conditional net factors are obtained as follows. At fixed $\mathbf{p}$, let $\bar{k}_e = \sum_{i\in e} p_i$. The four nonconstant expected energies of $(u_e, v_e) = (0,0), (1,0), (0,1), (1,1)$ are

$$\begin{aligned} \varepsilon_{e,00} &= 0, \\ \varepsilon_{e,10} &= w_e(r_e - 1 - \bar{k}_e) - \kappa_e, \\ \varepsilon_{e,01} &= w_e(\bar{k}_e - 1) - \kappa_e, \\ \varepsilon_{e,11} &= w_e(r_e - 2). \end{aligned} \tag{4}$$

Minimizing the corresponding energy and negative entropy gives $q_e(u,v) = \exp(-\varepsilon_{e,uv}/T)/Z_e$, where $Z_e = \sum_{u,v} \exp(-\varepsilon_{e,uv}/T)$. The reduced net free energy is $w_e - T \ln Z_e$. Differentiating this reduced function gives $w_e \langle v_e - u_e \rangle_{q_e}$ for every incident vertex. Thus the auxiliary distribution is updated exactly for the current vertex marginals, without storing an independent evolving auxiliary field or enumerating vertex assignments.

The net-auxiliary block produces a different interaction through its response to the vertex

marginals. Write $y_e = v_e - u_e$ and $\bar{k}_e = \sum_{i \in e} p_i$. Its state energies are affine in $\bar{k}_e$, and the derivative of each energy with respect to an incident p_i is $w_e y_e$. Differentiating the reduced log partition function gives $\partial f_e / \partial p_i = w_e \langle y_e \rangle$ and $\partial^2 f_e / \partial p_i \partial p_j = -w_e^2 \operatorname{Var}(y_e)/T$ whenever $i, j \in e$. This includes the diagonal because the optimal auxiliary factor responds when any incident vertex probability changes.

At the uncertain state, the auxiliary variance has an explicit form. Set $A_e = w_e(r_e/2 - 1)$. The nonconstant energies of states $(0,0), (1,0), (0,1), (1,1)$ are $0, A_e - \kappa_e, A_e - \kappa_e, 2A_e$. Symmetry gives $\langle y_e \rangle = 0$, while $Z_e = 1 + 2\exp[-(A_e - \kappa_e)/T] + \exp(-2A_e/T)$. Consequently

$$\chi_e = \operatorname{Var}(y_e) = \frac{1}{1 + \exp(-\kappa_e/T)\cosh(A_e/T)}. \tag{5}$$

Summing net responses gives the Hessian contribution

$$\mathbf{H}_{\text{net}} = -\mathbf{B}^{\mathsf{T}} \operatorname{diag}(w_e^2 \chi_e / T) \mathbf{B}. \tag{6}$$

Adding the vertex-entropy Hessian $4 \operatorname{diag}(T_i)$ and the exact balance Hessian $2\lambda(\mathbf{1}\mathbf{1}^{\mathsf{T}} - \mathbf{I})$ completes the local response before collective terms are included.

Constructing collective directions

Collective catalysts couple probabilities along extended directions that preserve mean block load to first order. We obtain these directions from the original incidence matrix $\mathbf{B}$, with one row per net and one column per vertex. For the unit-weight benchmark inputs, define $a_e = 1/\max(1, \lfloor r_e^2/4 \rfloor)$ and the implicit Laplacian

$$\begin{aligned} \mathbf{L} = {} & \operatorname{diag}\big(\mathbf{B}^{\mathsf{T}}(\mathbf{a} \odot \mathbf{r})\big) \\ & - \mathbf{B}^{\mathsf{T}} \operatorname{diag}(\mathbf{a}) \mathbf{B}. \end{aligned} \tag{7}$$

Its row sums vanish. Low-frequency directions therefore describe relative changes across the hypergraph, as in classical spectral partitioning[7,8]. Sparse incidence products apply this operator without storing its pairwise adjacency matrix.

Two geometries provide different candidate collective directions. The first uses $\mathbf{L}$ on vertices belonging to at least one net of size greater than one. For the second, let $\mathbf{D}_c = \operatorname{diag}((L_{ii})_i)$ on this cut-active support and let $\mathbf{P}_c$ remove the mean in each connected component. We use

$$\mathbf{L}_{\text{norm}} = \mathbf{P}_c \mathbf{D}_c^{-1/2} \mathbf{L} \mathbf{D}_c^{-1/2} \mathbf{P}_c. \tag{8}$$

These degree-normalized physical directions need not be eigenvectors of the unscaled operator. All resulting directions are expressed in vertex coordinates, made zero-mean and orthonormal, and padded with zeros outside the cut-active support. Excluding a vertex from this eigenproblem never removes it from the partitioning QUBO or its balance constraint.

Disconnected supports require both component contrasts and internal directions. We compute four directions for each geometry. On connected support, the constant direction is discarded. On disconnected support, random contrasts between components occupy the nonconstant nullspace. If the largest component contains more than half of all vertices, we retain one contrast and three internal directions; otherwise, up to four contrasts are used. A shift on the component nullspace isolates internal modes. The Lanczos calculation uses tolerance 10^{-6}, at most 20,000 iterations, subspace size 32 and probe seed 314159; accepted absolute residuals are below 10^{-5}.

Vertices with no cut response receive a separate temporary collective term because they still contribute to the capacity constraint. If $f > 1$ vertices are incident only to singleton nets or to no nets, a random subset of size $\lfloor f/2 \rfloor$ receives amplitude $1/\lfloor f/2 \rfloor$ and its complement receives $-1/\lceil f/2 \rceil$. Normalizing gives a zero-mean unit direction on these vertices. We use the main-text collective quadratic form to define C_{null}, with a separate envelope. One inactive vertex cannot support such a direction. The benchmark inputs *thermal1* and *ca-CondMat* have respectively 248 and 610 inactive vertices; these vertices remain in every FEM state.

Collective curvature

For the main-text collective interaction, take one unit-norm mode $\mathbf{v}$ with $\mathbf{1}^{\mathsf{T}}\mathbf{v} = 0$. Writing $m_i = 2p_i - 1$, independence gives $\mathbb{E}[(\mathbf{v}^{\mathsf{T}}\mathbf{s})^2] = (\mathbf{v}^{\mathsf{T}}\mathbf{m})^2 + \sum_i v_i^2(1 - m_i^2)$. The binary self-interaction subtracts $\sum_i v_i^2$, leaving the following expectation

and its probability gradient:

$$\mathbb{E}_q[C_v] = -\frac{\rho}{2}\left[(\mathbf{v}^\mathsf{T}\mathbf{m})^2 - \sum_i v_i^2 m_i^2\right], \qquad (9)$$
$$\nabla_p \mathbb{E}_q[C_v] = -2\rho\left[\mathbf{v}(\mathbf{v}^\mathsf{T}\mathbf{m}) - \mathbf{v}^{\odot 2}\odot\mathbf{m}\right].$$

For independent Bernoulli vertices, removing diagonal self-interactions in the collective expectation gives exactly $-4\rho(\mathbf{v}\mathbf{v}^\mathsf{T} - \mathrm{diag}(v_i^2))$ as its probability Hessian. The Rayleigh value along a unit mode is $-4\rho(1-\sum_i v_i^4)$. Because $\mathbf{1}^\mathsf{T}\mathbf{v} = 0$, the mean-balance rank-one term contributes zero in this direction; the balance-variance diagonal term remains in the full Hessian. The argument identifies a local balanced instability, not a path through all near-optimal partitions, a quantum spectral gap or an OGP remedy. For nonnegative unit-sum mode weights, the weighted operator is the sum of these exact quadratic Hessians. Null-support directions obey the same identity.

Variational landscape

The two-mode slice in Fig. S1 isolates the local mechanism near the uncertain state. The minimum projected curvature is 2.67 without catalysts and −1.30 with catalysts: a local bowl develops an unstable direction. This does not measure an overlap gap or a global energy barrier.

The supplementary variational slice uses $\mathbf{p} = \mathbf{1}/2 + q_1\mathbf{v}_1 + q_2\mathbf{v}_2$; zero-mean orthonormal modes keep the mean balance fixed. Vertex entropy, the full balance expectation and diagonal corrections to the collective expectation are retained.

The slice illustrates the local curvature; the update calculation below tests whether that curvature predicts the dynamics used by the solver.

From free-energy curvature to the actual update

The curvature statement can be connected to the implemented dynamics without identifying FEM with a physical quantum evolution. We freeze all temperatures, penalties and catalyst coefficients and consider the interior stationary state $\mathbf{p}_\star = \mathbf{1}/2$, with count target $n/2$. This includes even strict instances and the symmetric odd-count encoding. The oriented odd-count penalty generally makes this state nonstationary, so the following linearization does not apply there. Field clipping must be inactive in a neighbourhood of the stationary state. Let $\phi(\mathbf{p})$ denote the reduced interaction free energy after conditional auxiliary minimization, excluding vertex entropy and balance, and put $\mathbf{G} = \nabla_p^2\phi(\mathbf{p}_\star)$. The complete probability Hessian is

$$\mathbf{H} = \mathbf{G} + 4\,\mathrm{diag}(T_i) + 2\lambda(\mathbf{1}\mathbf{1}^\mathsf{T} - \mathbf{I}). \qquad (10)$$

This retains the balance-variance term. Replacing it by only a squared mean-count error would give a different Hessian and a different update.

Proposition 1 (frozen-update linear stability). For $\lambda \geq 0$, positive step sizes d_i and temperatures T_i, define $r_i = d_i/(1+2\lambda d_i)$, $k_i = r_i/(1+r_iT_i)$, $\mathbf{K} = \mathrm{diag}(k_i)$ and

$$\mathbf{M} = \left(\mathbf{K}^{-1} + (\lambda/2)\mathbf{1}\mathbf{1}^\mathsf{T}\right)^{-1}. \qquad (11)$$

The exact implicit-root update has field Jacobian

$$\mathbf{J} = \mathbf{I} - \tfrac{1}{4}\mathbf{M}\mathbf{H}. \qquad (12)$$

Consequently, negative curvature of $\mathbf{H}$ implies a real multiplier of $\mathbf{J}$ greater than one. Strict linear stability holds precisely when every eigenvalue of $\mathbf{M}^{1/2}\mathbf{H}\mathbf{M}^{1/2}$ lies in $(0, 8)$. Positive curvature alone does not exclude a discrete-step overshoot instability.

Proof. For a field perturbation $\mathbf{h}$, sigmoid expansion gives $\delta\mathbf{p} = \mathbf{h}/4$, magnetization $\mathbf{m} = \mathbf{h}/2$, and sparse-gradient perturbation $\delta\mathbf{g} = \mathbf{G}\mathbf{h}/4$. Linearizing the tilted field in Methods gives $\widetilde{\mathbf{h}} = [\mathbf{I} - \mathbf{K}\,\mathrm{diag}(T_i) + (\lambda/2)\mathbf{K} - \mathbf{K}\mathbf{G}/4]\mathbf{h}$. The implicit root gives $\delta = \mathbf{1}^\mathsf{T}\mathbf{h}'/4$ and therefore $[\mathbf{I} + (\lambda/2)\mathbf{K}\mathbf{1}\mathbf{1}^\mathsf{T}]\mathbf{h}' = \widetilde{\mathbf{h}}$. Subtracting the same equation evaluated at $\mathbf{h}$ yields Equation (12). The mobility is positive definite. Thus $\mathbf{J}$ is similar to $\mathbf{I} - \mathbf{M}^{1/2}\mathbf{H}\mathbf{M}^{1/2}/4$. Congruence preserves the numbers of positive and negative eigenvalues, which proves both assertions. This is a local result for frozen coefficients and an accurately solved count root; it is not a convergence result for a changing annealing path.

A negative Rayleigh quotient on a zero-mean vector is sufficient for an unstable mode of the full update. With heterogeneous mobility, that unstable eigenvector need not have exactly zero count component. A balanced-subspace eigenvalue is an exact multiplier prediction only when the subspace is invariant. The regular construction below

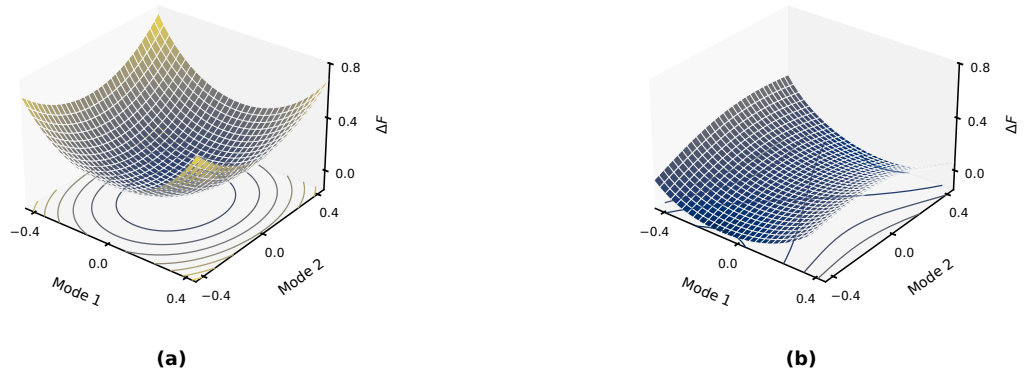


Figure S1 Variational slice; (a) Uncatalyzed and **(b)** catalyzed free energy on example B at temperature 1 and penalty 0.01, using the same two lowest nonconstant incidence modes; The catalyst has pair strength 1.3, auxiliary scale 1 and collective strength 1 with equal mode weights; Each surface shows $F(\mathbf{p}) - F(\mathbf{1}/2)$ on identical axes and color scales; every p_i lies between 0.05 and 0.95; This is a classical variational projection

satisfies this condition and tests the prediction without that ambiguity.

A critical catalyst strength from incidence

A threshold follows without choosing a successful binary partition. Fix temperature, penalty and auxiliary polarization, and let $\mathbf{U}$ have orthonormal columns spanning the zero-mean subspace. Suppose $\mathbf{H}_{0B} = \mathbf{U}^\mathsf{T}\mathbf{H}_0\mathbf{U}$ is positive definite without the pair catalyst, and write $\mathbf{A}_B = \mathbf{U}^\mathsf{T}\mathbf{A}_{\text{off}}\mathbf{U}$. The balanced Hessian with pair strength $\gamma \geq 0$ is $\mathbf{H}_{0B} - 2\gamma\mathbf{A}_B$.

Proposition 2 (incidence-dependent threshold). Put $g_{\max} = \lambda_{\max}(\mathbf{H}_{0B}^{-1/2}\mathbf{A}_B\mathbf{H}_{0B}^{-1/2})$. If $g_{\max} > 0$, the first loss of positive definiteness occurs at

$$\gamma_c = \frac{1}{2g_{\max}}. \tag{13}$$

If $g_{\max} \leq 0$, a nonnegative pair strength cannot destabilize this initially positive balanced Hessian.

Proof. Congruence by $\mathbf{H}_{0B}^{-1/2}$ converts the Hessian to $\mathbf{I} - 2\gamma\mathbf{H}_{0B}^{-1/2}\mathbf{A}_B\mathbf{H}_{0B}^{-1/2}$. Its first zero eigenvalue has the stated value. This criterion distinguishes structural couplings that can destabilize an uncertain balanced direction from those that cannot. It predicts local response, not whether the resulting direction reaches a good cut.

An analytic hypergraph and numerical checks

For an even cyclic 3-uniform hypergraph, take $e_i = \{i, i+1, i+2\}$ modulo n. Every vertex has incidence degree three. With unit weights, $a_e = 1/2$ and uniform auxiliary strength κ, the nonconstant Fourier modes have $\mathbf{B}^\mathsf{T}\mathbf{B}$ eigenvalues $\beta_k = [1 + 2\cos(2\pi k/n)]^2$, $k = 1, \ldots, n-1$. Their probability curvatures are

$$\begin{aligned} h_k &= 4T - 2\lambda - \frac{\chi(T)\beta_k}{T} - \gamma(\beta_k - 3), \\ \chi(T) &= [1 + e^{-\kappa/T}\cosh(1/(2T))]^{-1}. \end{aligned} \tag{14}$$

Here κ is the actual auxiliary energy coefficient; the numerical auxiliary-strength setting multiplies $r_e/2 - 1 = 1/2$. For uniform d, let $k = d/(1 + 2\lambda d + dT)$. Balanced modes are invariant under both the Hessian and mobility, so their actual update multipliers are $j_k = 1 - kh_k/4$. For the tested $n = 12$ and 24, $\beta_\star = [1 + 2\cos(2\pi/n)]^2 > 3$, the pair threshold at an initially stable temperature is

$$\gamma_c(T) = \frac{4T - 2\lambda - \chi(T)\beta_\star/T}{\beta_\star - 3}. \tag{15}$$

Thus catalyst strength can be chosen to cross a specified local stability boundary rather than selected solely from a final-cut scan.

A contrasting cyclic hypergraph on four vertices has the same net size and incidence degree, both three, but every nonconstant incidence eigenvalue is $\beta_k = 1$. Its pair operator on the balanced

subspace is $-\mathbf{I}$, so the pair catalyst adds $2\gamma\mathbf{I}$ to that Hessian. Hence it stabilizes these directions and cannot induce a curvature instability from an initially positive balanced Hessian. Automatic derivatives and production-update finite differences at the three plotted strengths verify this control as well. Degree and net size alone therefore do not determine the sign of the catalytic response.

We verify these expressions on $n = 12$ and 24, temperatures 0.5, 1.5 and 4, pair strengths 0, 1.3 and 4, and auxiliary-strength settings 0 and 1. Across all 36 FP32 configurations, automatic probability Hessians agree with the analytic matrix within 1.9×10^{-6}, and finite differences of the production implicit update agree with Equation (12) within 7.1×10^{-7}. Sixteen further checks on irregular 8- and 10-vertex hypergraphs use unequal incidence degrees, vertex step sizes and local temperatures. Their non-symmetric update Jacobians agree with the mobility formula within 2.9×10^{-7}, and a negative Hessian eigenvalue gives a multiplier above one as predicted. Eight additional threshold checks at the initially stable temperatures 1.5 and 4 agree between the generalized-eigenvalue and Fourier formulas; perturbing the predicted strength by relative $\pm 10^{-4}$ changes the curvature sign as predicted.

Figure 2(d)–(f) uses $\lambda = 0.01$, $d = 0.125$ and no auxiliary polarization for the plotted cyclic controls. On cooling $n = 12$ from $T = 6$ to 0.25, the first curvature zeros are 0.93489, 1.93100 and 4.66839 for the three pair strengths. At $T = 1.5$, a 10^{-7} field perturbation in the first nonconstant Fourier mode decays or grows with predicted multipliers 0.90652, 1.05892 and 1.37544. Thirty-two actual nonlinear updates agree with the linear amplitude prediction to relative error below 1.8×10^{-6}. These are controlled small-field tests at fixed coefficients. They test a local growth law rather than measuring a transition between globally good partitions.

Scope of the local FEM mechanism

The curvature and update calculations describe a neighbourhood of an uncertain stationary state. A global overlap-gap claim asks a different question about the whole low-cost set[15,16]. For threshold τ, define $\mathcal{S}_\tau = \{\mathbf{x} : \sum_i x_i = b,\ E_{\mathrm{cut}}(\mathbf{x}) \le \tau\}$ and complement-invariant overlap $q(\mathbf{x}, \mathbf{y}) = |n^{-1} \sum_i (2x_i - 1)(2y_i - 1)|$. An overlap gap requires an interval of overlaps absent between the appropriate low-cost states, with the ensemble and threshold specified. The present real-graph runs do not establish that property. A catalyst also need not evade a restriction on a specified stable algorithm class. Our mechanism claims are confined to local free-energy and update stability; final partition quality is evaluated separately.

Supplementary Note 4. Shared FEM solver policy

Primary FP32 single-trajectory comparison

The current main tables and time–quality figures use one complete cohort: 30 original inputs, three capacities and three new trials per solver and setting. FEM evolves a single FP32 trajectory and then applies the conditional-QUBO CPU refiner specified in Methods. Profile selection is unchanged from the earlier latency control below and precedes these runs. No current evaluation cut selects a profile, and no historical winner is substituted into the primary tables. This is a study of previously calibrated inputs, not unseen-instance policy generalization.

The temperature and penalty schedules remain those of the historical profiles below, rescaled to a maximum of 3,000 plus 144 updates. Strict balance uses the complement-symmetric encoding; bands use bounded binary slack. Count roots use 64 safeguarded iterations. The plateau rule below is unchanged and all temporary terms vanish in the terminal original-QUBO phase. The CPU input reader stores sparse incidence; its gain-based refiner conditionally minimizes the exact QUBO over net auxiliaries. It uses eight passes, eight candidates on each side, a 512-move plateau cap and a 0.5 s improvement budget after initialization and capacity repair. Feasible pair moves and within-band single moves are scored on the original cut nets, and only the best prefix is retained.

All 270 final FEM partitions are feasible. Eleven strict trials lack a feasible raw checkpoint and are repaired explicitly; they remain in all primary summaries. Independent NumPy rescoring checks 12,681 saved states, reconstructs initial fields and verifies frozen input, mode,

source and library hashes. The primary KaHyPar-K cohort comprises 270 independently rescored direct single-call outputs on the local CPU. It has no imposed historical 64- or 128-second repeated-search budget. The limitations of direct exact-balance calls are discussed in Supplementary Note 1.

Quality and timing are paired by trial. Table 2 of the main text uses the best cut among each solver's three trials; Table 3 uses the median individual process time from those same runs. Neither table claims that the median time produced the best-of-three cut. The process clock includes CPU refinement but excludes prior mode construction and calibration. Initial-label refinement controls, independent audits and archival are also outside the timed solve. Complete trials, raw cuts, selected checkpoints, stopping outcomes and stage times accompany the released data. Reference targets for the main figures use only the best of the three KaHyPar-K calls, rather than a selected historical FEM cut. All 30 inputs receive equal weight separately at each capacity; missing incumbents contribute zero quality until feasibility is observed.

Table S1 separates the strict-capacity raw trajectory, its refined endpoint and the same CPU refinement applied directly to random initial labels. Each column is best of the same three seeds; raw feasibility is the count of feasible trajectories. The random-label control tests the contribution of the discrete stage without an anneal, not a time-matched state-of-the-art partitioner. Its audit cost is not charged to the FEM solve. Historical multi-replica measurements below retain their original precision, seeds and budgets and are not pooled with these current results.

The FEM objective, conditional updates and implicit count equation are specified in Methods. This note supplies the numerical replica profiles and discrete schedules needed to reproduce them, followed by complete group outcomes, capacity checks and the original and follow-up results. The policy follows mean-field annealing[17] and the FEM framework[18]; primary and exploratory budgets remain separate throughout.

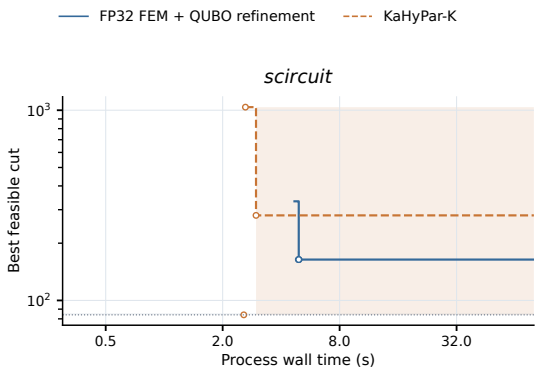


Figure S2 Primary FP32 strict-capacity progress on *scircuit*; Median and full range of three trials; process clock, endpoint refinement and reference definition match Fig. 4 of the main text.

FEM replica allocation

Each 16-replica group starts from a shared Gaussian field of standard deviation 0.01 plus independent replica perturbations of standard deviation 0.001. The same 16 fields are copied across the three groups.

The common 48-replica policy assigns 16 replicas to local catalysts, 16 to unscaled modes and 16 to normalized modes. The latter groups cycle through first mode, second mode, equal first two and normalized inverse-eigenvalue weights over all four. The inverse uses floor $\max(10^{-12}, 10^{-6}\max_\ell \mu_\ell)$ to handle null modes. These choices cross with strengths $\{0.5, 1, 2, 4\}\max(1, \bar{D})$. The local, auxiliary and per-spin profiles act simultaneously in every group. All three groups receive identical nearby initial fields. Temperature annealing follows the classical free-energy search principle[19]; the trajectories evolve independently. Exchange Monte Carlo[20] is a different mechanism and is not used by this shared policy.

The same 16 local profiles are used in every geometry group, with all their numerical parameters given in Table S2. For profile a, α and ξ define its pair-node scale $z_i = (1+D_i)^{-\alpha}\exp(\xi\sigma_i^{\text{contrast}})$. The temperature signal and ζ define $T_i = T\exp(\omega\zeta\sigma_i)$; "none" means $\zeta = 0$. The pair and auxiliary amplitudes multiply their common envelopes. In a collective group, rows 1–4, 5–8, 9–12 and 13–16 receive strengths 0.5, 1, 2 and

Table S1 Primary FP32 strict-capacity readout controls; Best native cut over three trials, with the same seeds in all columns. A dash indicates no feasible raw FEM checkpoint. Refined columns include explicit capacity repair where needed; no input or failed raw trial is removed.

Graph	Raw FEM	FEM + refinement	Random + refinement	Raw feasible
IBM01	416	383	1,891	3/3
IBM02	451	448	1,063	3/3
IBM03	1,284	1,240	4,260	3/3
IBM04	982	909	4,829	3/3
IBM05	2,886	2,402	6,790	3/3
IBM06	1,087	1,043	5,651	3/3
IBM07	1,300	1,239	7,200	3/3
IBM08	1,268	1,226	8,548	3/3
IBM09	755	708	10,029	3/3
IBM10	1,962	1,801	11,115	3/3
IBM11	1,461	1,392	12,859	3/3
IBM12	2,977	2,836	14,495	3/3
IBM13	2,572	2,507	15,862	3/3
IBM14	3,391	2,517	27,428	3/3
IBM15	5,061	4,873	35,823	3/3
IBM16	5,798	5,623	44,878	2/3
IBM17	3,784	3,541	60,393	3/3
IBM18	–	1,857	53,428	0/3
2cubes_sphere	4,156	4,049	99,781	3/3
G2_circuit	–	9,022	8,744	0/3
PGPgiantcompo	494	477	1,435	2/3
airfoil_2d	467	352	541	3/3
bcsstk29	372	372	13,615	3/3
ca-CondMat	4,548	4,372	7,817	3/3
cfd1	2,982	2,645	69,678	3/3
finan512	146	146	24,442	3/3
parabolic_fem	1,026	1,026	382,261	3/3
scircuit	332	164	34,001	3/3
shallow_water2	–	2,515	3,132	0/3
thermal1	263	256	9,853	3/3

4, respectively. Within each four-row block, the choices are the first mode, second mode, equal first two modes, and inverse-eigenvalue weighting over all four. This specifies how local profiles and collective choices are paired, rather than leaving their joint allocation implicit.

Profile 14 implements its geometry transition by mixing two fixed pair gradients. The mixing weight is zero through annealing fraction 0.25, rises as a squared sine to one at fraction 0.70, and stays one thereafter. Thus the two endpoint geometries have $\alpha = 0$ and $\alpha = 0.5$, each with its own matched total coupling mass. This is a convex combination of fixed quadratic interactions, rather than an interaction inferred from the evolving vertex labels.

FEM annealing and terminal schedules

The historical full-portfolio run uses $K = 12{,}000$ annealing updates and 144 terminal updates. For annealing index $j = 0, \ldots, K-1$, the common temperature is $T_j = 8(0.02/8)^{j/(K-1)}$. The penalty is 0.01 for $j < K - 1200$ and then follows $\lambda_j = 0.01(\lambda_{\text{fin}}/0.01)^{[j-(K-1200)]/1199}$. The terminal updates use $T = 0.02$, $\lambda = \lambda_{\text{fin}}$ and zero temporary interaction and temperature-offset coefficients.

All interaction envelopes are defined by the same smooth ramp. Write $R(\tau; a, b) = \sin^2[\frac{\pi}{2}\,\text{clip}((\tau - a)/(b - a), 0, 1)]$ for $\tau = j/(K-1)$. The pair and auxiliary envelope is $R(\tau; 0, 0.10)[1 - R(\tau; 0.75, 1)]$; the collective envelope is $R(\tau; 0, 0.15)[1 - R(\tau; 0.80, 1)]$; and the inactive-vertex envelope is $R(\tau; 0.55, 0.70)[1 - R(\tau; 0.96, 1)]$. Collective amplitudes multiply

Table S2 Complete local profiles within each FEM geometry group; Pair and auxiliary columns are envelope amplitudes. The auxiliary coefficient also multiplies $w_e \max(r_e/2 - 1, 0)$. All profiles are used on every input.

Profile	α	ξ	Temperature signal	ζ	Pair	Auxiliary	η_a
1	0	−0.75	Degree	1.5	1.30	0.40	1.0
2	0	−0.75	Degree	1.5	1.00	0.70	1.0
3	0	−0.75	Degree	1.5	1.15	0.55	1.0
4	0	−0.75	Degree	1.5	1.00	0.40	1.0
5	0	0	Degree	1.5	1.15	0.40	1.0
6	0	0	Degree	1.5	0.85	0.40	1.0
7	0	0	Degree	1.5	1.00	0.40	1.0
8	0	0	Degree	1.5	1.30	0.55	1.0
9	0	0.75	Degree	0.75	1.15	0.55	1.0
10	0	0.75	Degree	0.75	1.00	0.55	1.0
11	0	0	Contrast	−1.5	0.70	0.25	1.0
12	0	0	Contrast	−1.5	0.85	0.40	1.0
13	0.5	0	None	0	2.00	0	0.5
14	$0 \to 0.5$	0	None	0	2.20	0	0.5
15	0	0	None	0	1.00	0.55	1.0
16	0	0	None	0	1.15	0.70	1.0

$\max(1, \bar{D})$, whereas inactive-vertex amplitudes multiply λ_{fin}. Local-temperature offsets use $\omega(\tau) = \sin^2(\pi\tau/0.9)$ for $\tau < 0.9$ and zero otherwise. Each envelope is therefore zero at the annealing endpoints and remains zero in the terminal stage.

The initialization, checkpoint criterion and all-replica descent are specified in Methods. The tables below retain that original protocol; the symmetric and stopping follow-ups state their distinct readout and budget separately.

Performance of the three FEM geometry groups

The original 20-instance subset uses seed 21 and one strict 48-replica batch per graph. This policy was fixed before these outcomes, after previous studies had already evaluated every graph. Within each group, the common 16 profiles and strengths are unchanged. The three groups receive matched random fields and 12,144 updates. Their costs are charged as one batch; group quality columns below do not imply separate measured GPU times. All 960 trajectories have independently reconstructed initial fields, native cuts, full-QUBO scores, endpoint coefficients and binary states. No winning raw state needs a descent flip. Further exploratory controls are not pooled into that historical uniform policy.

Mode residuals, localization and component counts are in the compact study summaries. Normalization improves *IBM08* and *IBM15* but does not make the localized *ca-CondMat* directions useful. Its unscaled internal-mode inverse participation ratios are 0.99989, 0.99978 and 0.33310, compared with 0.08216, 0.05746 and 0.04777 after normalization. These remain localized directions: the catalytic curvature factor $1 - \sum_i v_i^4$ is nearly zero for the first two unscaled modes. Component contrasts and degree normalization alone do not provide a successful global direction.

Matched balance

The ten strict extension cases use the same 48 profiles, seed 21 and 12,144 updates as the original subset. Table S4 reports each group's feasible raw cut. The complete strict *scircuit* endpoint is 318 after six original-QUBO flips, compared with raw cut 373. The 3% *scircuit* solve failed both its original 24-iteration and 64-iteration residual checks before producing a complete result. Captured and uncaptured updates agree on the failing saturated field. The final retry uses 64 iterations of the same implicit equation, a central-quarter Newton acceptance bracket and the unchanged tolerance; both failed attempts (325.28 s in total) are charged separately. This numerical exception does not change the objective, initial fields or catalyst schedules. The separate 1% and 3% suites

Table S3 Portfolio groups; Native cut (seconds);

Graph	Local	Unscaled	Normalized	Spectral (s)	KaHyPar 0% (s)
IBM01	325	496	509	476 (0.62 s)	287 (128.68 s)
IBM02	497	477	513	625 (1.88 s)	367 (128.30 s)
IBM03	1,217	1,171	1,303	1,555 (1.25 s)	1,247 (128.61 s)
IBM04	621	674	667	929 (1.26 s)	696 (128.67 s)
IBM06	1,320	1,080	1,095	1,348 (1.34 s)	1,335 (129.22 s)
IBM08	1,577	1,248	1,238	2,103 (54.31 s)	1,200 (128.37 s)
IBM09	1,504	756	805	967 (6.38 s)	703 (128.67 s)
IBM12	2,486	2,763	2,443	4,082 (22.60 s)	3,138 (128.81 s)
IBM15	5,270	5,134	4,573	6,106 (57.95 s)	4,554 (65.63 s)
IBM17	3,838	3,446	3,778	4,623 (40.36 s)	3,390 (144.28 s)
IBM18	3,522	2,197	2,396	3,518 (62.99 s)	2,772 (130.88 s)
bcsstk29	732	372	372	426 (0.84 s)	360 (129.15 s)
2cubes_sphere	5,004	4,125	4,137	4,289 (10.09 s)	6,892 (129.14 s)
G2_circuit	3,071	624	624	624 (43.55 s)	1,144 (129.09 s)
PGPgiantcompo	463	463	466	540 (1.76 s)	396 (128.57 s)
finan512	299	146	146	146 (16.05 s)	169 (128.59 s)
thermal1	5,545	255	255	272 (30.52 s)	291 (128.47 s)
airfoil_2d	413	375	376	395 (0.78 s)	372 (128.35 s)
ca-CondMat	4,473	4,522	4,522	6,035 (6.18 s)	4,202 (128.85 s)
parabolic_fem	4,490	1,026	1,026	1,026 (1467.23 s)	1,256 (133.58 s)

use bounded binary slack, its conditional Gibbs distribution and the full count/slack variance in the expectation. Their vertex fields and catalyst schedules match the strict suite. No strict winner is substituted into a band result, and no classical partition enters initialization.

The 1% protocol specifies 64 central-safeguarded iterations for *scircuit* before observing its new batch, and 24 iterations for the other cases. All 30 batches complete without a numerical retry. Independent reconstruction checks all 1,440 trajectories and both raw and selected encoded states. KaHyPar uses seed 0, a nominal 64 s repeated partitioning budget and one thread per process; up to four reference processes execute concurrently. The corresponding recorded process time is separate from its nominal budget.

Raw cut is the best feasible field-sign output before descent. Valid counts denote raw replicas with zero auxiliary gap and the allowed count bound, out of 48. Flips refer to the replica selected after original-QUBO descent; that replica can differ from the best feasible raw replica. All original-policy solutions satisfy their configured bound. These counts and cut differences distinguish feasibility from objective quality.

Original uniform-policy benchmark

Table S6 retains every original 48-replica FP64 result before any follow-up selection. It supplies the quality values paired with the original charged times in the main text. The strict, 1% and 3% settings each use one independent batch of 12,144 updates.

Matched complement-symmetric follow-up

The follow-up covers every odd-sized input in the 30-instance set and *IBM01* as an even-sized control. Both encodings receive the same two independently generated seed batches, original-graph random fields, certified mode arrays, 48 profiles and 3,000 annealing plus 144 cold updates. Arithmetic is FP32 and both arms use 64 centrally safeguarded root iterations with the same precision-dependent residual check. The historical frozen benchmark remains FP64 with its own numerical protocol. These are distinct studies.

Checkpoints occur every 128 updates and at phase boundaries. Each arm uses the same readout: retain each replica's lowest native cut among unchanged vertex states satisfying $b \le N \le n - b$, then select the best replica. Both label orientations are accepted in both arms. Net auxiliary bits

Table S4 Strict extension groups; Native cut;

Graph	Local	Unscaled	Normalized	Spectral	KaHyPar 0%
IBM05	1,901	1,947	1,936	2,171	1,807
IBM07	1,361	1,163	1,256	1,869	1,428
IBM10	2,694	2,619	2,070	2,569	1,626
IBM11	1,632	1,581	1,642	1,958	1,671
IBM13	2,544	2,641	2,677	2,725	1,204
IBM14	3,606	3,491	3,259	4,663	2,273
IBM16	2,907	3,368	3,845	6,068	4,044
cfd1	2,064	2,206	2,206	3,420	3,121
shallow_water2	882	768	768	768	909
scircuit	2,014	373	401	215	68

Table S5 Raw feasibility and readout

	Raw valid / 48			0%		1%		3%	
Graph	0%	1%	3%	Raw cut	Flips	Raw cut	Flips	Raw cut	Flips
IBM01	47	48	48	325	0	368	0	335	0
IBM02	44	48	48	477	0	454	0	420	0
IBM03	47	48	48	1,171	0	1,160	0	1,143	0
IBM04	46	47	47	621	0	636	0	609	0
IBM05	45	48	48	1,901	0	1,896	0	1,895	0
IBM06	47	48	48	1,080	0	1,078	0	1,080	0
IBM07	48	48	48	1,163	0	1,144	0	1,206	0
IBM08	47	47	48	1,238	0	1,181	0	1,211	0
IBM09	48	48	48	756	0	723	0	685	0
IBM10	47	48	48	2,070	0	1,732	0	1,720	0
IBM11	46	48	48	1,581	0	1,458	0	1,417	0
IBM12	45	48	47	2,443	0	2,457	0	2,492	0
IBM13	48	48	47	2,544	0	2,508	0	1,921	0
IBM14	47	48	48	3,259	0	3,088	0	3,577	0
IBM15	48	48	48	4,573	0	4,721	0	4,145	0
IBM16	47	48	48	2,907	0	3,770	0	3,544	0
IBM17	48	48	48	3,446	0	3,366	0	2,784	0
IBM18	45	46	48	2,197	0	2,231	0	2,326	0
2cubes_sphere	48	48	48	4,125	0	3,991	0	3,951	0
G2_circuit	40	48	48	624	0	624	0	624	0
PGPgiantcompo	37	48	48	463	0	467	0	457	0
airfoil_2d	48	47	48	375	0	373	0	342	0
bcsstk29	47	42	34	372	0	360	0	360	0
ca-CondMat	39	48	48	4,473	0	4,406	0	4,215	0
cfd1	48	47	48	2,064	0	2,174	0	2,250	0
finan512	45	46	48	146	0	146	0	146	0
parabolic_fem	42	48	48	1,026	0	1,026	0	1,026	0
scircuit	32	44	48	373	6	277	0	46	0
shallow_water2	38	48	48	768	0	768	0	768	0
thermal1	37	48	48	255	0	257	0	257	0

Table S6 Original shared-policy cut and measured imbalance (%);

	0%		1%		3%	
Graph	FEM	KaHyPar	FEM	KaHyPar	FEM	KaHyPar
IBM01	325 (0.000)	**287** (0.000)	368 (0.988)	**220** (0.471)	335 (2.964)	**202** (2.760)
IBM02	477 (0.000)	**367** (0.000)	454 (0.990)	**350** (0.520)	420 (3.000)	**337** (2.989)
IBM03	**1,171** (0.000)	1,247 (0.000)	1,160 (0.994)	**962** (0.000)	1,143 (2.939)	**957** (0.199)
IBM04	**621** (0.000)	696 (0.000)	636 (0.996)	**598** (0.996)	609 (2.995)	**582** (1.898)
IBM05	1,901 (0.000)	**1,807** (0.000)	1,896 (0.402)	**1,725** (0.000)	1,895 (0.450)	**1,719** (1.036)
IBM06	**1,080** (0.000)	1,335 (0.000)	1,078 (0.560)	**1,023** (0.923)	1,080 (0.892)	**988** (1.994)
IBM07	**1,163** (0.000)	1,428 (0.000)	1,144 (0.997)	**948** (0.997)	1,206 (2.996)	**910** (2.992)
IBM08	1,238 (0.000)	**1,200** (0.000)	1,181 (0.998)	**1,140** (0.000)	1,211 (2.276)	**1,140** (0.000)
IBM09	756 (0.000)	**703** (0.000)	723 (0.996)	**627** (0.974)	685 (2.996)	**620** (2.663)
IBM10	2,070 (0.000)	**1,626** (0.000)	1,732 (0.746)	**1,340** (0.000)	1,720 (0.668)	**1,333** (0.349)
IBM11	**1,581** (0.000)	1,671 (0.000)	1,458 (0.998)	**1,079** (0.989)	1,417 (2.999)	**1,062** (2.891)
IBM12	**2,443** (0.000)	3,138 (0.000)	2,457 (0.788)	**1,996** (0.979)	2,492 (2.895)	**1,936** (2.476)
IBM13	2,544 (0.000)	**1,204** (0.000)	2,508 (1.000)	**848** (0.404)	1,921 (3.000)	**848** (0.404)
IBM14	3,259 (0.000)	**2,273** (0.000)	3,088 (1.000)	**1,849** (0.000)	3,577 (3.000)	**1,849** (0.000)
IBM15	4,573 (0.000)	**4,554** (0.000)	4,721 (0.999)	**2,839** (0.722)	4,145 (0.586)	**2,741** (1.868)
IBM16	**2,907** (0.000)	4,044 (0.000)	3,770 (1.000)	**2,117** (0.832)	3,544 (2.746)	**1,969** (2.941)
IBM17	3,446 (0.000)	**3,390** (0.000)	3,366 (0.999)	**2,351** (0.965)	2,784 (3.000)	**2,310** (2.600)
IBM18	**2,197** (0.000)	2,772 (0.000)	2,231 (0.537)	**1,961** (0.000)	2,326 (2.989)	**1,915** (1.355)
2cubes_sphere	**4,125** (0.000)	6,892 (0.000)	3,991 (0.989)	**3,922** (0.993)	3,951 (2.999)	**3,884** (2.993)
G2_circuit	**624** (0.000)	1,144 (0.000)	**624** (0.779)	**624** (0.000)	**624** (0.779)	**624** (0.000)
PGPgiantcompo	463 (0.000)	**396** (0.000)	467 (0.449)	**352** (0.000)	457 (2.959)	**350** (0.000)
airfoil_2d	375 (0.000)	**372** (0.000)	373 (0.999)	**328** (0.000)	342 (2.997)	**324** (2.899)
bcsstk29	372 (0.000)	**360** (0.000)	**360** (0.172)	**360** (0.000)	**360** (0.172)	**360** (0.000)
ca-CondMat	4,473 (0.000)	**4,202** (0.000)	4,406 (0.994)	**3,918** (0.916)	4,215 (3.000)	**3,855** (3.000)
cfd1	**2,064** (0.000)	3,121 (0.000)	2,174 (0.999)	**1,564** (0.725)	2,250 (2.998)	**1,564** (0.725)
finan512	**146** (0.000)	169 (0.000)	**146** (0.000)	**146** (0.000)	**146** (0.000)	**146** (0.000)
parabolic_fem	**1,026** (0.000)	1,256 (0.000)	**1,026** (0.013)	**1,026** (0.000)	**1,026** (0.013)	**1,026** (0.000)
scircuit	318 (0.000)	**68** (0.000)	277 (0.999)	**46** (0.513)	46 (2.451)	**42** (2.273)
shallow_water2	**768** (0.000)	909 (0.000)	768 (0.764)	**640** (0.000)	768 (0.601)	**640** (0.000)
thermal1	**255** (0.000)	291 (0.000)	257 (0.102)	**239** (0.111)	257 (0.099)	**239** (0.114)

are minimized exactly with those vertices fixed; no vertex descent, contraction, external initializer or capacity repair is used. Thus a gain cannot be attributed merely to accepting an opposite block label at output.

Table S7 includes every outcome. The main table selects the minimum verified strict cut over the original policy and all four follow-up runs on each tested input; superscript f marks a new winner. This is an explicitly selected envelope with additional computation, not the outcome of a newly fixed policy. Its band columns retain their original independent results. Search seconds below sum both encodings and both seeds, including checkpoints. They exclude cached-mode construction, state archival and independent auditing. Setup and the historical mode charges remain in the machine-readable records; these search times do not replace the original pipeline timings.

Independent NumPy scoring checks 7,488 complete binary states from 2,496 trajectories: the minimum-energy, common-capacity and final-checkpoint states of all 52 runs. These are three saved views of each trajectory, not independent repetitions. Every reported winner satisfies strict integer capacity and has zero auxiliary gap. Initial fields are reconstructed from each seed; paired fields and mode hashes match. All three saved state sets are bitwise identical between encodings on *IBM01*. The original and symmetric penalties differ on odd inputs through a linear field, so these on-only catalytic comparisons do not separate encoding effects from their interaction with catalysts.

Table S7 Complete strict-capacity follow-up; Native cut for both seed batches. Best is the minimum over these four runs; all outputs have 0% observed imbalance. Time sums all four search loops.

	Oriented		Symmetric				
Graph	Seed A	Seed B	Seed A	Seed B	Best	KaHyPar	Time (s)
IBM01	311	295	311	295	295	287	13.00
IBM02	484	484	484	484	484	367	15.62
IBM04	672	672	671	671	671	696	20.62
IBM05	1,954	1,954	1,979	1,984	1,954	1,807	18.35
IBM08	1,268	1,268	1,268	1,268	1,268	1,200	33.13
IBM09	756	756	755	755	755	703	29.57
IBM10	2,098	2,098	2,096	2,096	2,096	1,626	37.59
IBM13	2,553	2,536	1,165	1,165	1,165	1,204	50.90
IBM14	3,560	3,243	2,808	3,243	2,808	2,273	87.16
IBM17	3,636	3,636	3,634	3,634	3,634	3,390	110.14
IBM18	1,884	1,884	1,928	1,900	1,884	2,772	123.30
ca-CondMat	4,362	4,555	4,389	4,581	4,362	4,202	19.49
parabolic_fem	1,026	1,026	1,026	1,026	1,026	1,256	364.87

Time–quality trajectories and a matched stopping control

This diagnostic measures progress within the search rather than the cost of the complete pipeline. We use the preselected weak inputs *IBM02*, *IBM10* and *scircuit*, all three allowed imbalances, and two independently seeded batches at each setting. Each batch has 48 replicas in the same three groups and uses FP32, 64 safeguarded count-root iterations, 3,000 annealing updates and at most 144 terminal updates. The strict encoding is centered and complement-symmetric; the band uses bounded binary slack. Initial Gaussian draws are shared across the three groups, and each fixed and adaptive pair starts from identical fields and cached mode arrays. Checkpoint selection uses the best native cut among feasible unchanged vertex labels, with no vertex repair or descent. This budget is distinct from the historical 12,000-update FP64 policy.

The recorded trajectory is the best feasible native cut seen by each checkpoint. Each readout checks the integer count and scores the native nets; an infeasible checkpoint cannot improve the incumbent. The archived FEM-only plot holds the previous checkpoint value between measurements and shows the median and full range of both seeds, without averaging across capacities. The synchronized search clock includes checkpoints but starts after mode loading and setup and ends before serialization. Saved raw, capacity-selected and final endpoint states are independently rescored and initial fields are reconstructed. Intermediate vertex states are not retained, so the trajectory between saved endpoints is supported by the recorded cut/count trace rather than independent binary replay.

The adaptive continuation tests a conservative observable plateau. After at least 1,024 updates, the best feasible native cut must be unchanged for 512 updates. The current checkpoint must contain at least four feasible replicas and a replica attaining the incumbent with complement-invariant label turnover at most 0.001 and mean active-vertex probability variance at most 0.02. Passing these criteria switches to the original endpoint energy with zero catalysts and terminal temperature. At least 64 terminal updates and two consecutive stable terminal checkpoints are required before stopping. The fixed control continues to the full 3,144-update budget, while the adaptive run may also exhaust that budget. These observables do not certify optimality; the matched continuation checks for quality loss in this finite cohort.

Table S8 lists every fixed and adaptive outcome, including cases that use the full budget. These data measure observed search savings; neither the plateau criterion nor these finite continuations certify optimality or universal quality preservation.

Table S8 All matched stopping outcomes; Every seed and capacity of the three-input diagnostic is shown; time is the synchronized search loop in seconds, excluding mode construction, setup and serialization.

	Allowed	Seed	Native cut		Updates		Search time (s)	
Input	imbalance (%)	batch	Fixed	Adaptive	Fixed	Adaptive	Fixed	Adaptive
IBM02	0	0	484	484	3144	3064	4.026	3.872
IBM02	0	1	484	484	3144	3064	4.017	3.866
IBM02	1	0	443	443	3144	3064	4.096	4.007
IBM02	1	1	443	443	3144	3064	4.065	3.986
IBM02	3	0	398	398	3144	3144	4.143	4.086
IBM02	3	1	398	398	3144	3144	4.067	4.127
IBM10	0	0	2096	2096	3144	3064	9.366	9.106
IBM10	0	1	2096	2096	3144	3064	9.357	9.247
IBM10	1	0	1938	1938	3144	3064	9.491	9.270
IBM10	1	1	1938	1938	3144	3064	9.508	9.400
IBM10	3	0	1819	1819	3144	3064	9.481	9.249
IBM10	3	1	1819	1819	3144	3064	9.624	9.389
scircuit	0	0	421	421	3144	3064	25.882	25.200
scircuit	0	1	421	421	3144	3064	25.920	25.250
scircuit	1	0	361	361	3144	3064	26.188	25.527
scircuit	1	1	361	361	3144	3064	26.198	25.504
scircuit	3	0	44	44	3144	1216	26.275	10.165
scircuit	3	1	44	44	3144	1216	26.245	10.157

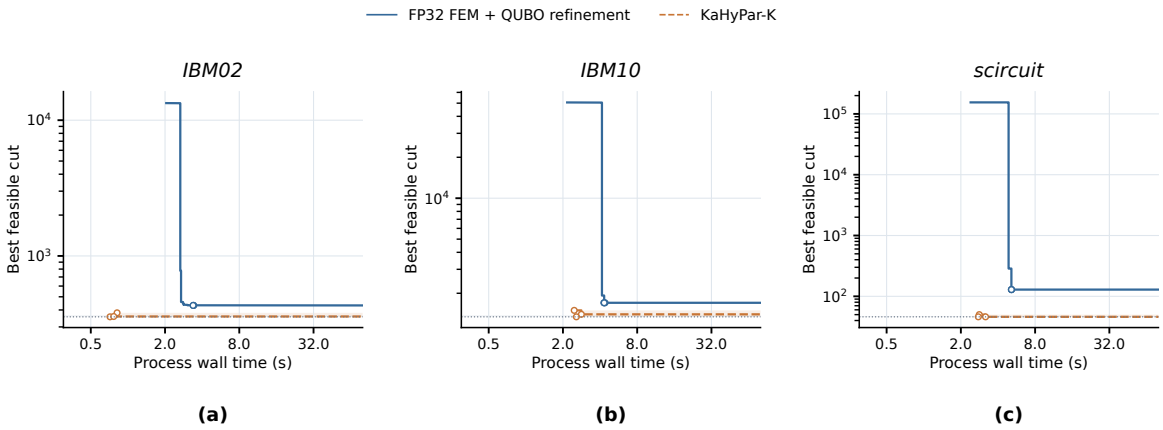


Figure S3 Primary FP32 priority-input progress at 1% allowed imbalance; (a) *IBM02*; **(b)** *IBM10*; **(c)** *scircuit*; all three trials per solver are retained, with the primary process clocks and conditional-QUBO refinement of the main text. References are the best of three KaHyPar-K calls.

Historical single-strategy latency control

This latency study evaluates all 30 primary inputs at each allowed imbalance, using one frozen strategy and three new random seeds per setting. The three priority inputs retain the earlier channel-control selection: calibration minimizes feasible cut on one seed, with ties broken by fewer effective channels and then a fixed order; a second seed checks that choice. On the remaining 27 inputs, the raw feasible winner of the corresponding historical 12,144-update FP64 run with 48 trajectories fixes the profile and geometry, retaining all catalyst channels. This uses no vertex descent and no new evaluation outcome to select a policy. It is an instance-specific latency diagnostic on previously studied inputs. The preceding

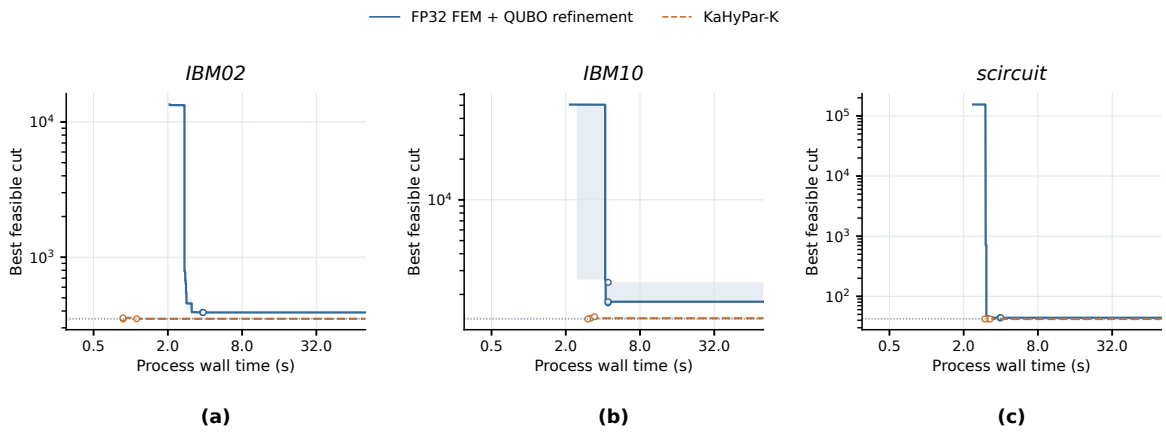


Figure S4 Primary FP32 priority-input progress at 3% allowed imbalance; (a) *IBM02*; **(b)** *IBM10*; **(c)** *scircuit*; all three trials per solver are retained, with the primary process clocks and conditional-QUBO refinement of the main text. References are the best of three KaHyPar-K calls.

48-trajectory stopping study is a separate cohort, and the two cohorts are not pooled.

For each priority input, calibration across the three capacities examines 33 previous 48-trajectory runs. Their search times sum to 134.09 s for *IBM02*, 314.66 s for *IBM10* and 863.53 s for *scircuit*. The three chosen-policy validation runs per input add 12.17, 28.60 and 78.61 s, respectively. These search-only costs exclude mode construction and process setup and are not included in the new single-trajectory latency curves. Thus those curves cannot be interpreted as first-use costs of choosing an instance-specific policy.

The selected policies use the profiles in Table S2. For *IBM02*, 0% and 1% use local profile 7 with pair and local-temperature terms; 3% uses local profile 2 with pair terms alone. For *IBM10*, 0% and 1% use normalized-mode profile 16 with collective terms alone, and 3% uses that profile with pair and collective terms. For *scircuit*, 0% and 1% use unscaled-mode profile 7 with collective terms alone; 3% uses unscaled-mode profile 3 with collective terms alone. The original temperature, penalty and catalyst schedules are retained up to an optional early transition to the terminal phase. Each solve has a single trajectory and 64 safeguarded count-root iterations per update, with at most 3,000 annealing and 144 terminal updates. No vertex repair or descent is applied. Arithmetic for closed catalyst channels is omitted, with matched checks against the unpruned update. The selected profiles for all 90 settings and all calibration search costs accompany the numerical data.

Early stopping requires both a native-cut plateau and a stable, polarized feasible partition. Checks occur every 64 annealing updates. A transition is permitted after at least 1,024 executed updates and 512 updates without an improved feasible cut. At that checkpoint the current cut must equal the best observed feasible cut. The fraction of changed cut-active vertex labels, minimized over exchanging the block names, must be at most 0.001, and their mean probability variance must be at most 0.02. These thresholds are fixed before the new trials and do not depend on a KaHyPar cut or reference target.

The terminal verification executes the original QUBO rather than stopping while temporary interactions remain active. It restores the final penalty, sets temperature to 0.02 and every temporary coefficient to zero. After an early transition, checks occur every 32 updates; runs that reach the scheduled boundary retain the saved 64-update grid and phase boundaries. Termination requires at least 64 terminal updates and two consecutive feasible, stable checks without a further best-cut improvement. A new improvement or unstable check resets this terminal patience; otherwise execution continues to the maximum budget. The retained binary incumbent is feasible when first observed. Neither this stopping rule nor a terminal plateau proves optimality.

A matched fixed-budget control covers all 27 new priority-input trials, rather than only favorable seeds. Every binary checkpoint shared before an early transition is unchanged, and all 27 final best feasible cuts match the fixed-budget continuation. The largest median search-time savings occur for *IBM02* at 1% and *scircuit* at 3%, with factors 1.42 and 1.68, respectively. The corresponding standalone process medians fall from 4.22 to 3.68 s and from 6.88 to 5.80 s; startup and setup remain charged. Most other settings remain near the maximum budget; this finite control does not establish quality preservation on every input.

FEM process time starts before launching the Python interpreter and includes imports, input, model and schedule setup, loading cached certified modes, graph capture, GPU search and host readout. Independent binary auditing and output archiving are outside the clock. Construction of the reused modes and historical calibration are excluded, so this is a cached-direction latency measurement rather than complete cold preprocessing. FEM uses an A100 GPU. KaHyPar-K uses one thread on the local CPU, with one cold native CLI call per trial, using all three seeds of the complete single-invocation benchmark. Its clock runs from CLI launch through the returned partition file. These observations upper-bound an unobserved internal target-discovery time. No pre-output incumbent is reconstructed. Every saved FEM checkpoint and every returned KaHyPar partition are independently rescored. The complete comparison has 270 FEM and 270 KaHyPar trials; stopping can change the number of FEM checkpoints. Every instance, seed and capacity is retained, including censored target failures. At 0%, 11 of 90 FEM trials have no feasible binary incumbent: all three trials on *IBM18*, *G2_circuit* and *shallow_water2*, and one trial each on *IBM16* and *PGPgiantcompo*. All eleven execute the maximum 3,144 updates. All 180 relaxed-capacity trials are feasible. These outcomes characterize this short-budget, historically calibrated policy; they are retained in the averages and do not isolate a precision effect.

The main figures use uniform references from the earlier main benchmark. For each capacity, C_{ref} is the smaller of its reported FEM and KaHyPar cuts, including the existing best-observed strict envelope. No new latency-trial cut enters a reference; these are attainable benchmark cuts rather than proved optima. This common historical-benchmark analysis is separate from the original narrower calibration-target analysis, which remains in the data. A reference may become smaller as capacity is relaxed, so reaching a fixed percentage of it need not be an easier absolute cut target.

Table S9 lists the priority-input references and every new cut. The data provide every trial for all 30 inputs; Table S10 summarizes their process times. A separate aggregate is constructed for each capacity: trials are averaged within an instance, then all 30 instances receive equal weight. Capacities are not pooled. Zero attained quality before a feasible output preserves missing feasibility coverage. Raw cuts from different-sized inputs are not averaged. Additional first-passage records use the same strict-capacity benchmark target at all three capacities, separating a looser constraint from a changed quality target.

For a common per-trial wall budget B, let $\hat{p}$ be the observed fraction reaching a fixed target by B. An independent-restart projection for reliability P is

$$T_P^{\mathrm{proj}} = B \left\lceil \frac{\log(1-P)}{\log(1-\hat{p})} \right\rceil . \tag{16}$$

At least one run is charged; zero observed hits give no finite plug-in projection. Budgets 1, 4, 8 and 16 s and reliabilities 0.97 and 0.99 are retained in the data. Wilson 95% probability intervals propagate to a range of restart projections. These are fixed-budget estimates, not observed first-hit times or demonstrated reliability guarantees; three trials per setting give wide uncertainty. A trial that ends without a hit retains its censored observation.

Supplementary Note 5. Mechanism controls

This note retains controls that distinguish the principal ingredients of the method: catalytic interactions, collective and inactive-vertex directions, structural weighting, and the timing of couplings versus local temperatures. All are exploratory comparisons on previously seen inputs. Their own seeds, replica counts and schedules are specified below; their best cuts are not substituted into the primary table. Earlier portfolio tuning, colder and longer continuations, pair-factor refreshes, replica mixtures and feedback

Table S9 All priority-input trials in the single-strategy latency comparison; Cuts are listed in the fixed evaluation-trial order; process times are medians of three trials in seconds, with the cached-mode clock defined above; references come from the earlier main benchmark, not these trials.

Input	Allowed (%)	Reference	FEM cuts	KaHyPar-K cuts	FEM (s)	KaHyPar-K (s)
IBM02	0	367	451, 451, 451	411, 500, 564	4.166	0.515
IBM02	1	350	433, 433, 433	356, 381, 358	3.675	0.766
IBM02	3	337	391, 391, 391	350, 349, 356	4.109	0.867
IBM10	0	1626	1962, 1962, 1962	2127, 2281, 2982	4.964	2.118
IBM10	1	1340	1924, 1924, 1924	1400, 1342, 1497	5.052	2.569
IBM10	3	1333	1764, 2688, 1792	1333, 1369, 1317	5.195	3.178
scircuit	0	68	332, 332, 332	1038, 84, 280	6.767	2.619
scircuit	1	46	286, 286, 286	46, 50, 46	6.936	2.820
scircuit	3	42	44, 44, 44	42, 42, 42	5.797	3.121

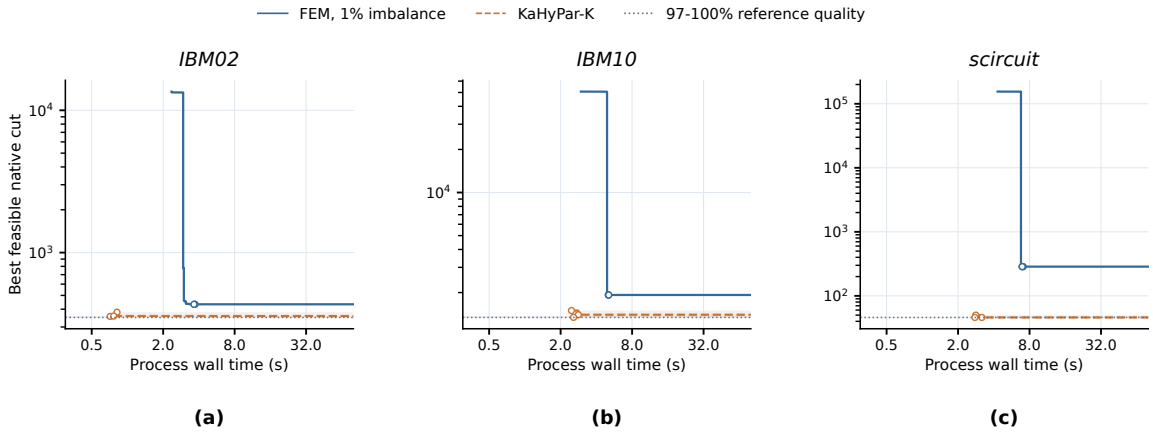


Figure S5 Native-cut progress at 1% allowed imbalance; (a) *IBM02*; **(b)** *IBM10*; **(c)** *scircuit*; each solver has three independent trials; lines show the median and shading the complete trial range; open circles mark final outputs; process clocks, cached modes, reference bands and observation rules follow the strict-capacity comparison in the main text; this figure contains only 1% trials.

trials are retained in the repository's historical control archive.

Catalyst-off and catalyst-on searches

A matched control switches the temporary vertex interactions and local temperatures off or on. Both arms use 48 replicas, seed 21, identical initial fields, FP64, 12,000 annealing plus 144 terminal updates, and 64 centrally safeguarded count-root iterations. Auxiliary polarization is absent in both arms. Final energy, capacity and original-QUBO descent are unchanged. Thus this control evaluates a specified vertex-catalyst combination rather than isolating every catalyst in the primary policy. The mixed arm spends the same 48 trajectories across four timing choices; it is retained as a regression control, not a best-of-policies result. Certified modes are shared; *rajat26* uses a diagnostic GPU construction and is outside the primary 30-input study.

Collective directions and inactive vertices

The resolved-mode study uses two seed batches, 15 and 16, with 16 replicas per arm and 12,144 updates. Every arm retains the same local portfolio and random fields; certified modes replace the earlier uncertified diffusion probes. The signed/hold arm additionally reverses the local pair sign early and changes the temperature schedule, so it does not isolate the collective term

Table S10 Process times for all 30 inputs; Medians of three trials in seconds; FEM uses observable early stopping and cached modes, KaHyPar-K uses one cold native call per trial; the process clocks include startup, setup and returned output; historical calibration and mode construction are excluded from FEM latency; [f] marks a setting with at least one trial without a feasible binary output, whose consumed time is retained.

	0% imbalance		1% imbalance		3% imbalance	
Input	FEM	KaHyPar-K	FEM	KaHyPar-K	FEM	KaHyPar-K
IBM02	4.166	0.515	3.675	0.766	4.109	0.867
IBM10	4.964	2.118	5.052	2.569	5.195	3.178
scircuit	6.767	2.619	6.936	2.820	5.797	3.121
IBM01	4.023	0.266	4.114	0.365	4.115	0.415
IBM03	4.389	0.515	4.466	0.716	4.506	0.866
IBM04	4.515	0.666	4.672	0.766	4.573	1.016
IBM06	4.419	0.918	4.418	1.166	4.785	1.367
IBM08	5.127	1.467	5.216	2.118	4.498	2.520
IBM09	4.836	1.769	4.941	1.567	5.085	1.718
IBM12	5.252	2.118	5.388	3.021	5.329	3.171
IBM15	7.061	7.329	7.354	13.146	7.402	17.366
IBM17	7.332	7.980	7.776	9.233	7.431	9.983
IBM18	7.585[f]	7.878	7.773	10.636	8.095	13.893
bcsstk29	4.432	0.164	3.233	0.214	3.225	0.265
2cubes_sphere	6.470	2.620	7.065	8.434	6.847	68.882
G2_circuit	6.680[f]	5.930	5.018	6.379	5.489	6.830
PGPgiantcompo	3.651[f]	0.164	4.141	0.264	4.285	0.265
finan512	3.905	1.117	3.972	1.217	4.024	1.818
thermal1	5.826	0.866	4.434	1.517	4.444	4.222
airfoil_2d	4.202	0.264	4.288	0.365	4.274	0.515
ca-CondMat	4.474	1.367	4.854	1.768	4.537	5.825
parabolic_fem	11.672	8.481	13.267	11.485	13.161	30.219
IBM05	4.320	0.866	4.454	1.116	4.441	1.317
IBM07	4.623	1.116	4.748	1.769	4.835	2.871
IBM11	5.157	2.169	5.510	2.469	5.342	2.821
IBM13	5.531	3.020	5.606	3.822	5.816	3.621
IBM14	6.416	5.575	6.702	7.981	6.718	9.234
IBM16	7.208[f]	11.623	7.745	15.614	7.550	17.456
cfd1	6.365	2.368	6.112	3.571	6.406	9.783
shallow_water2	5.195[f]	1.317	4.014	1.668	3.956	2.969

Table S11 QUBO cut and measured imbalance;

Graph	Off	Shared catalyst	Mixed catalyst	KaHyPar
finan512	516 (0.000)	**146** (0.000)	**146** (0.000)	169 (0.000)
IBM13	5,150 (0.000)	2,444 (0.000)	1,941 (0.000)	**1,204** (0.000)
rajat26	722 (0.000)	209 (0.000)	228 (0.000)	**32** (0.000)

alone. Auxiliary-minimized readout and complete-QUBO descent are shown separately below.

The modes-only arm has no feasible raw output on *thermal1*. Adding a temporary direction on its 248 inactive vertices restores a feasible raw cut; these vertices remain in the original capacity constraint. The table retains this failed endpoint and the competing schedule choices. Component localization and mode certificates are assessed independently in Supplementary Notes 3 and 7.

Catalysis versus direct spectral readout

A separate mixed-mode control uses the same four certified directions for temporary couplings and

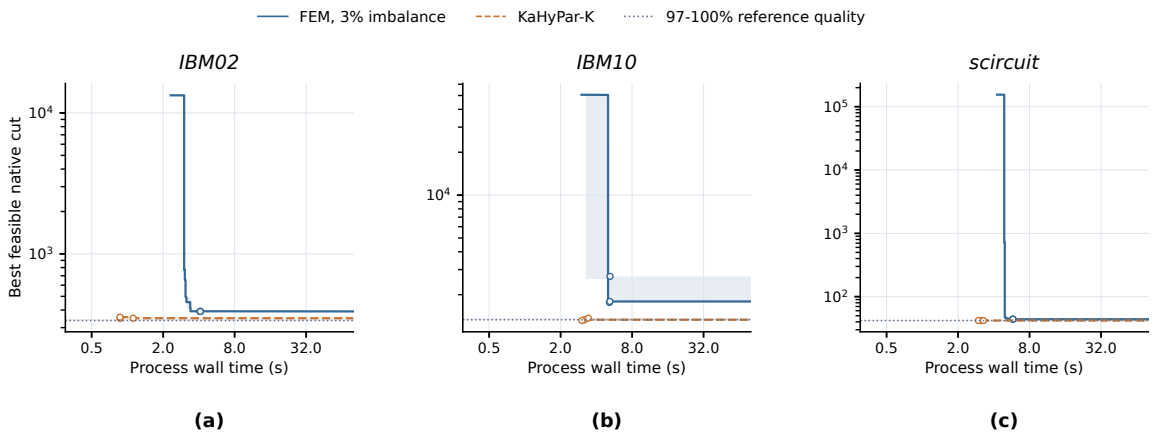


Figure S6 Native-cut progress at 3% allowed imbalance; (a) *IBM02*; **(b)** *IBM10*; **(c)** *scircuit*; conventions and three-trial sampling match Fig. S5; this figure contains only 3% trials, retaining the full seed-dependent range.

Table S12 Collective ablations; Median cut (seconds); a dash denotes no feasible raw output;

Graph	Arm	Raw	Selected	Strict KaHyPar	KaHyPar 3%
finan512	Long control	610 (26.01 s)	610 (26.71 s)	169 (128.59 s)	146 (64.52 s)
finan512	Signed/hold	621.5 (26.01 s)	528 (26.70 s)	169 (128.59 s)	146 (64.52 s)
finan512	Modes	146 (26.54 s)	146 (27.20 s)	169 (128.59 s)	146 (64.52 s)
finan512	Modes/hold/null	146 (26.58 s)	146 (27.28 s)	169 (128.59 s)	146 (64.52 s)
thermal1	Long control	5,546.5 (25.15 s)	2,301.5 (25.95 s)	291 (128.47 s)	239 (66.02 s)
thermal1	Signed/hold	3,248.5 (24.86 s)	2,000.5 (25.64 s)	291 (128.47 s)	239 (66.02 s)
thermal1	Modes	–	276 (26.82 s)	291 (128.47 s)	239 (66.02 s)
thermal1	Modes/null	255 (26.76 s)	255 (27.52 s)	291 (128.47 s)	239 (66.02 s)
thermal1	Modes/hold/null	264 (26.86 s)	264 (27.63 s)	291 (128.47 s)	239 (66.02 s)

for classical balanced sorting. The binary spectral outputs never initialize FEM. The table compares the complete charged pipelines and retains two independent strict KaHyPar references. Spectral rounding matches the catalytic cut on *finan512* and *G2_circuit* at lower cost. On *thermal1*, catalysis improves cut 272 to 255, below the two strict KaHyPar cuts 291 and 303. On *IBM02*, cut 477 improves on rounding's 775 but remains above KaHyPar's 367 and 364. These are four seen-input controls; their budgets differ from the primary study.

Structural weighting and schedule attribution

Five matched 16-replica paths on six seen inputs use seed 21, identical initial fields, 12,000 annealing plus 144 terminal updates. A rotation changes the basis within the same four-mode span. Coherence weighting redistributes auxiliary strength with net alignment to that span while preserving its total mass; leverage heating modifies the temporary local temperature by mode participation. Combining all three does not improve any of the six controls. Full definitions and all numerical outcomes remain in the data archive; these comparisons identify sensitivity to the chosen directions and weights, rather than a universal best rule.

The timing control then changes either coupling envelopes, local temperature envelopes, or both, while retaining global temperature, capacity penalty, initial fields, mode arrays and 48-replica budgets. Four arms on two seen inputs give 384 trajectories, all independently rescored. This separates the two timing channels: the larger improvement on *IBM13* comes from local-temperature timing, while every retimed arm worsens *rajat26*. The schedules are classical variational controls, distinct from an inhomogeneous quantum driver[21].

Table S13 Collective and classical references; Native cut (charged seconds); both strict KaHyPar seeds are retained;

Graph	Mixed raw	Spectral	KaHyPar strict 0	KaHyPar strict 1	KaHyPar 3%
finan512	146 (37.67 s)	146 (6.26 s)	169 (128.59 s)	184 (257.22 s)	146 (64.52 s)
thermal1	255 (41.53 s)	272 (9.94 s)	291 (128.47 s)	303 (257.13 s)	239 (66.02 s)
IBM02	477 (16.42 s)	775 (1.61 s)	367 (128.30 s)	364 (128.40 s)	337 (64.44 s)
G2_circuit	624 (62.08 s)	624 (15.20 s)	1,144 (129.09 s)	1,274 (129.40 s)	624 (67.27 s)

Table S14 Structural controls; Native cut (charged seconds); Bold marks row minima for cut and time;

Graph	Control	Rotation	Coherence	Heating	Combined	KaHyPar 0%
IBM02	457 (18.31 s)	476 (14.53 s)	443 (**14.37** s)	560 (14.50 s)	495 (14.51 s)	**367** (128.30 s)
IBM08	1,248 (78.37 s)	1,413 (**74.05** s)	1,235 (74.12 s)	1,432 (74.09 s)	1,500 (74.17 s)	**1,200** (128.37 s)
IBM09	739 (35.18 s)	1,719 (27.20 s)	758 (**27.10** s)	711 (27.21 s)	833 (27.26 s)	**703** (128.67 s)
bcsstk29	402 (26.02 s)	402 (17.55 s)	402 (17.49 s)	402 (**17.45** s)	402 (17.55 s)	**360** (129.15 s)
PGPgiantcompo	465 (20.20 s)	470 (12.01 s)	466 (12.08 s)	492 (**11.96** s)	497 (11.98 s)	**396** (128.57 s)
ca-CondMat	4,543 (23.74 s)	4,580 (19.26 s)	4,543 (19.08 s)	4,529 (19.25 s)	4,562 (**19.04** s)	**4,202** (128.85 s)

Table S15 Timing attribution; Native cuts; all partitions have 0% observed imbalance; bold marks the better matched cut;

Graph	Original	Coupling only	Temperature only	Both	KaHyPar
IBM13	2,544	2,514	2,233	2,211	**1,204**
rajat26	170	202	193	202	**32**

Supplementary Note 6. Computational cost

Timing is separated into complete charged pipelines and diagnostic components. The primary tables below retain every mode, setup, search, checkpoint and readout charge. The kernel and precision controls use their own numerical paths; they do not replace those historical costs. Additional scaling, repeated profiling and alternative sparse-action measurements are retained in the data archive.

Primary timing and hardware

The complete pipeline time charges CPU construction of both mode geometries, input/graph setup, compilation and capture, FEM checkpoints and all 48 single-bit descents. The two CPU jobs can overlap, so their sum is a conservative charged cost rather than a parallel critical-path measurement. Reference times retain their original CPU execution and balance setting. The primary KaHyPar references use one thread of an Intel Core i7-12700. The kernel-fusion, penalty-feedback and single-precision controls use an Intel Xeon E5-2620 v3 CPU for their references. Primary reference runs allow at most two concurrent CPU processes. GPU search uses an NVIDIA A100 with 40 GB memory. These hardware differences are retained when interpreting the measured times.

Search bottlenecks and fused kernels

The implementation uses Torch[22], CUDA execution graphs[23] and sparse incidence products[24]. An instrumented *IBM01* replay contains 25,824 kernel calls. Device kernels occupy 164.44 ms of a 189.76 ms window. Count-root kernels account for 35.1% of device time, sparse products for 24.6%, and other kernels for 40.2%. These fractions refer to that window, not the complete pipeline or every input.

The fused control combines custom dense kernels, grouped sparse products, 32-update execution graphs and 64 safeguarded root iterations. It retains 48 replicas, FP64, seed 21 and 12,144 updates. GPU LOBPCG with component deflation and polishing constructs the modes[25,26]. Of 30 strict-capacity inputs, 29 pass the fixed mode certificate. On *scircuit*, residual 5.38×10^{-4} exceeds the 10^{-5} limit and its 11.99 s failed

Table S16 Pipeline timing; Seconds;

Graph	Modes	Graph	Runner	FEM	Descent	Total
IBM01	0.61	0.78	10.62	13.05	0.29	25.35
IBM02	1.86	0.55	10.75	17.82	0.43	31.42
IBM03	1.23	0.71	10.89	19.84	0.51	33.19
IBM04	1.23	0.68	10.82	21.57	0.61	34.91
IBM06	1.31	0.71	10.80	23.79	0.70	37.31
IBM08	54.26	0.91	11.26	38.17	1.08	105.68
IBM09	6.33	0.98	10.72	40.34	1.37	59.74
IBM12	22.54	1.29	10.90	54.92	1.58	91.24
IBM15	57.80	3.16	11.43	132.51	4.07	208.97
IBM17	40.18	3.09	11.74	155.57	4.94	215.52
IBM18	62.79	2.72	12.00	169.73	5.53	252.76
bcsstk29	0.83	0.66	10.23	20.06	0.69	32.47
2cubes_sphere	9.99	2.18	11.44	105.52	3.80	132.94
G2_circuit	43.46	2.00	11.47	103.10	3.48	163.52
PGPgiantcompo	1.75	0.46	10.37	12.04	0.25	24.87
finan512	16.01	1.23	11.12	56.25	1.84	86.46
thermal1	30.45	1.57	11.55	61.08	2.27	106.93
airfoil_2d	0.76	0.81	10.31	16.01	0.44	28.33
ca-CondMat	6.17	0.64	11.35	22.23	0.61	40.99
parabolic_fem	1466.81	6.91	13.10	467.52	17.31	1971.66

Table S17 Strict extension timing; Seconds;

Graph	Modes	Graph	Runner	FEM	Descent	Total
IBM05	0.84	0.87	10.63	22.77	0.64	35.75
IBM07	4.02	0.88	11.02	32.86	1.00	49.79
IBM10	18.50	1.14	11.21	51.79	1.55	84.20
IBM11	11.24	1.31	10.70	53.63	1.76	78.65
IBM13	24.93	1.41	10.97	63.78	2.18	103.27
IBM14	80.66	2.04	11.45	114.51	3.52	212.18
IBM16	126.68	2.73	11.41	152.33	4.61	297.77
cfd1	8.34	1.57	11.07	70.01	3.31	94.29
shallow_water2	10.43	1.19	11.14	49.92	1.73	74.41
scircuit	3205.45	2.56	11.89	127.70	4.58	3352.19

construction is retained. Successful runs charge 186.15 s of mode construction, 33.58 s of setup, 3,252.37 s of search and 82.38 s of readout. Search is 91.5% of their 3,554.48 s total; the following examples separate search from pipeline cost.

Cost of the relaxed-capacity update

The band penalty introduces work that is absent from the fixed-target strict update. Its conditional slack mean sums over $U-L+1$ distinct integer values at every FEM step, including their encoding multiplicities. Increasing the allowed imbalance enlarges this range; it does not reduce a fixed number of updates. This implementation cost is separate from the fact that a looser capacity admits every strictly feasible partition.

A matched-policy component measurement uses the same local profile, random field, 64 root iterations and execution graphs at all three capacities. Across the three priority inputs, relaxed-capacity updates cost about 4–6% more GPU time per update than strict updates. The conditional slack target alone costs 10–18 μs in these probes. These measurements use a separate GPU and exclude checkpoints and process startup; they diagnose the additional arithmetic and do not

Table S18 3% pipeline timing; Seconds;

Graph	Modes	Graph	Runner	FEM	Descent	Failed	Total
IBM01	0.61	0.75	6.29	13.24	1.09	0.00	21.98
IBM02	1.86	0.65	6.75	18.18	2.02	0.00	29.46
IBM03	1.23	0.63	5.96	20.07	1.90	0.00	29.79
IBM04	1.23	0.90	6.71	21.81	2.35	0.00	33.00
IBM05	0.84	0.65	6.30	23.19	2.46	0.00	33.43
IBM06	1.31	0.71	6.25	24.31	2.74	0.00	35.32
IBM07	4.02	0.86	7.28	33.64	4.08	0.00	49.88
IBM08	54.26	0.88	6.96	39.05	4.55	0.00	105.71
IBM09	6.33	0.98	6.48	40.94	4.54	0.00	59.28
IBM10	18.50	1.31	6.83	52.70	5.98	0.00	85.31
IBM11	11.24	1.19	6.41	54.42	7.56	0.00	80.82
IBM12	22.54	1.18	6.61	55.45	5.95	0.00	91.72
IBM13	24.93	1.65	7.06	65.75	7.17	0.00	106.56
IBM14	80.66	2.02	6.57	117.32	13.44	0.00	220.02
IBM15	57.80	2.43	6.78	133.91	15.89	0.00	216.81
IBM16	126.68	3.14	6.74	154.13	16.63	0.00	307.32
IBM17	40.18	2.62	6.70	157.22	17.03	0.00	223.75
IBM18	62.79	2.69	6.96	171.77	19.64	0.00	263.85
2cubes_sphere	9.99	1.94	7.05	105.88	9.93	0.00	134.80
G2_circuit	43.46	2.77	7.51	106.69	12.98	0.00	173.42
PGPgiantcompo	1.75	0.70	5.93	12.33	0.90	0.00	21.61
airfoil_2d	0.76	0.64	5.82	16.25	1.34	0.00	24.81
bcsstk29	0.83	0.66	5.83	20.23	1.91	0.00	29.45
ca-CondMat	6.17	0.62	6.27	22.38	2.08	0.00	37.52
cfd1	8.34	1.65	6.61	70.75	7.78	0.00	95.12
finan512	16.01	1.49	6.63	57.45	6.72	0.00	88.30
parabolic_fem	1466.81	7.09	7.95	471.56	54.50	0.00	2007.91
scircuit	3205.45	2.67	5.70	190.23	15.25	325.28	3744.58
shallow_water2	10.43	1.47	6.26	50.11	7.04	0.00	75.30
thermal1	30.45	1.34	6.54	61.33	7.36	0.00	107.03

Table S19 Fused-kernel time (s);

Graph	Modes	Search	Total	KaHyPar
IBM01	6.33	19.73	**27.01**	128.82
IBM18	9.13	254.25	272.33	**131.20**
finan512	6.45	83.54	**93.37**	129.01
thermal1	9.20	97.00	**109.79**	129.81
parabolic_fem	40.47	714.09	778.44	**131.67**

replace the A100 process clocks in the main comparison. The measured cost need not rise monotonically from 1% to 3%. Observable early stopping addresses the number of executed updates, while optimizing this conditional target addresses their individual cost.

Standalone initialization and resident requests

A separate nine-setting probe separates repeated-request latency from standalone startup. Reusing one input, capacity and policy preserves all binary checkpoints for the matched random seed while retaining independent new fields for subsequent requests. On a separate GPU, *IBM02* at 0% changes from 4.70 s for an instrumented standalone process to 1.04 s for the matched resident request; *scircuit* at 3% changes from 6.21 to 0.83 s. Dependency imports cost about 1.2–1.4 s; *scircuit*'s model and incidence construction adds about 1.6–2.0 s. Reusing the CUDA context, incidence, schedules and captured execution graphs removes their repetition, but requires a prepared

Table S20 Precision-control cut and imbalance (%); QUBO on *IBM01*;

Allowed (%)	FP64	FP32	KaHyPar
0	325 (0.000)	438 (0.000)	**318** (0.000)
1	311 (0.988)	368 (0.988)	**211** (0.988)
3	314 (2.651)	284 (2.102)	**202** (2.760)

Table S21 Precision-control time (s); QUBO on *IBM01*;

Allowed (%)	FP64	FP32	KaHyPar
0	27.01	**21.66**	128.82
1	27.76	**22.32**	64.47
3	27.82	**22.39**	64.30

service for the same problem. These are cache-reuse measurements rather than a C++ implementation comparison; the main A100 curves retain standalone process costs.

Initial flat portions can also contain search before binary feasibility, beyond initialization. In the A100 strict-capacity *IBM02* trials, preparation takes about 2.32 s and the first feasible binary partition appears after a further 1.74–1.79 s of search. Accelerating startup therefore addresses only part of the observed delay; an earlier feasible readout also requires a change in the search dynamics that preserves final cut quality.

Single-precision control

The precision control repeats *IBM01* at 0%, 1% and 3% imbalance in FP32, with TensorFloat-32 disabled. Integer cut and capacity checks are unchanged. Independent FP64 calculations check roots, auxiliary responses and interaction gradients. The continuous root tolerance is $\max(10^{-4}, 16\epsilon_{\mathrm{mach}} n)$ in FP32 and $\max(10^{-7}, 10^{-11} n)$ in FP64; binary feasibility remains exact. All 288 saved raw/selected states pass independent scoring, and the largest measured FP32 root residual is 1.22×10^{-4}.

The paired solver runs reuse identical certified FP64 mode arrays; their construction charge remains 6.33 s. A separate FP32 sparse-Lanczos mode diagnostic, with FP64 projection and correction, costs 3.96 s and passes the same residual and orthogonality certificates. This construction saving is not substituted into the solver comparison.

Cuts change in both directions, so this single-input control does not estimate a general precision effect. Search accelerates more than the charged pipeline because setup and preprocessing remain outside the accelerated loop. The diagnostic is separate from both the primary FP64 benchmark and the later short-budget FP32 studies.

Certified mode construction

Sparse incidence actions apply the operators in Supplementary Note 3 without an expanded clique. CPU Lanczos[27] is checked against the original operator, with absolute residual below 10^{-5} and orthogonality and balance errors below 10^{-8}. The two mode geometries have separate costs. On *parabolic_fem*, mode construction dominates the primary pipeline, as reported in the main text. This motivates distinguishing construction, loading cached arrays, and the repeated FEM search.

For *scircuit*, the original unscaled shifted-Lanczos construction was stopped after 2776.60 s without a completed mode set. A component-deflated, Jacobi-preconditioned LOBPCG block solve[25] on the same incidence operator took 163.77 s on CPU; a further 11.37 s of CuPy refinement on A100 certified all four directions. CuPy[26] supplies the GPU array and sparse-linear-algebra backend; the original operator residual, rather than solver termination alone, is the certificate. Independent full-operator residuals are below 10^{-10}; balance and orthogonality errors are below 3×10^{-11} and 6×10^{-14}. This alternative was specified before evaluating partition quality on this instance. It supplies continuous coefficient modes, never a binary partition. The charged time includes the unsuccessful CPU attempt, CPU warm start, GPU refinement and separate normalized construction. The warm refinement time is not a standalone GPU construction time. A separate CuPy solve from random continuous vectors takes 19.08 s and independently reaches residuals below 10^{-10}. It is an acceleration diagnostic and does not replace the primary charged timing.

Supplementary Note 7. Validation and scope

Independent validation separates correctness of the objective and readout from claims about search quality and generality. We reconstruct initial fields, rescore complete binary states and verify integer capacities. The netlists and matrix-derived inputs come from the KaHyPar benchmark collection[28] and the University of Florida sparse matrix collection[29]. The control studies in this Note are assessed separately from the 30-input primary benchmark.

Endpoint validation

Independent scoring recomputes native cut, complete QUBO energy, integer block loads and auxiliary gap from the original input and saved binary labels. A feasible score requires the requested capacity and zero gap between encoded net cost and native cut. Initial fields are reconstructed from the recorded seeds, and endpoint coefficients are checked to have full penalty and zero catalysts. Numerical root accuracy is assessed separately from binary feasibility.

The primary studies audit all 4,320 trajectories. The strict-capacity follow-up checks 7,488 saved binary views of 2,496 trajectories from 52 runs; views of one trajectory are not independent repetitions. Original-QUBO descent accepts improving individual-bit flips and has a 50,000-flip limit. Incomplete states remain recorded with their stop reason and are excluded from feasible scores. Every benchmark reference is independently rescored under the same integer capacities.

For two seeds, a median is their average and can be fractional; geometric means use per-input median ratios. Such small seed samples do not establish population confidence intervals or robust time-to-target estimates. The repository archive retains the larger audit inventory for historical controls omitted from this supplement.

The stopping diagnostic adds 36 runs and 1,728 trajectories. Independent NumPy scoring checks their 5,184 saved raw, capacity-selected and final binary states. Reconstructed initial fields and paired hashes verify that the fixed and adaptive continuations share the same random start, mode arrays and numerical library. Endpoint selection is checked against the recorded native-cut/count traces; intermediate continuous fields are outside this replay audit. The complete 18-pair results appear in Supplementary Note 4.

Spectral validation

The resolved-mode, transfer, subspace and mixed controls contribute 288, 128, 32 and 128 trajectories, respectively: 576 new independently scored trajectories. Saved initial fields are reconstructed from seeds; all initial, raw and selected complete states are checked for native cut, QUBO cost, auxiliary gap and balance. Final penalty is $D_{\max} + 1$ and all catalysts are zero after 12,144 updates. Manual gradients agree with automatic derivatives, and GPU execution reproduces the direct updates for single-mode and subspace terms. All four additional KaHyPar states and all 16 classical spectral partitions are independently rescored. These checks verify the numerical construction and returned states; they do not certify global optimality.

Scope of the controls

The optimizer, temperature-schedule, degree and contrast controls test specific mechanisms on their stated inputs. The retained controls use previously seen inputs. Historical policy-selection and held-out designations remain documented in the repository archive. Reference partitions remain outside FEM initialization, and failed outputs remain part of the reported scope.

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